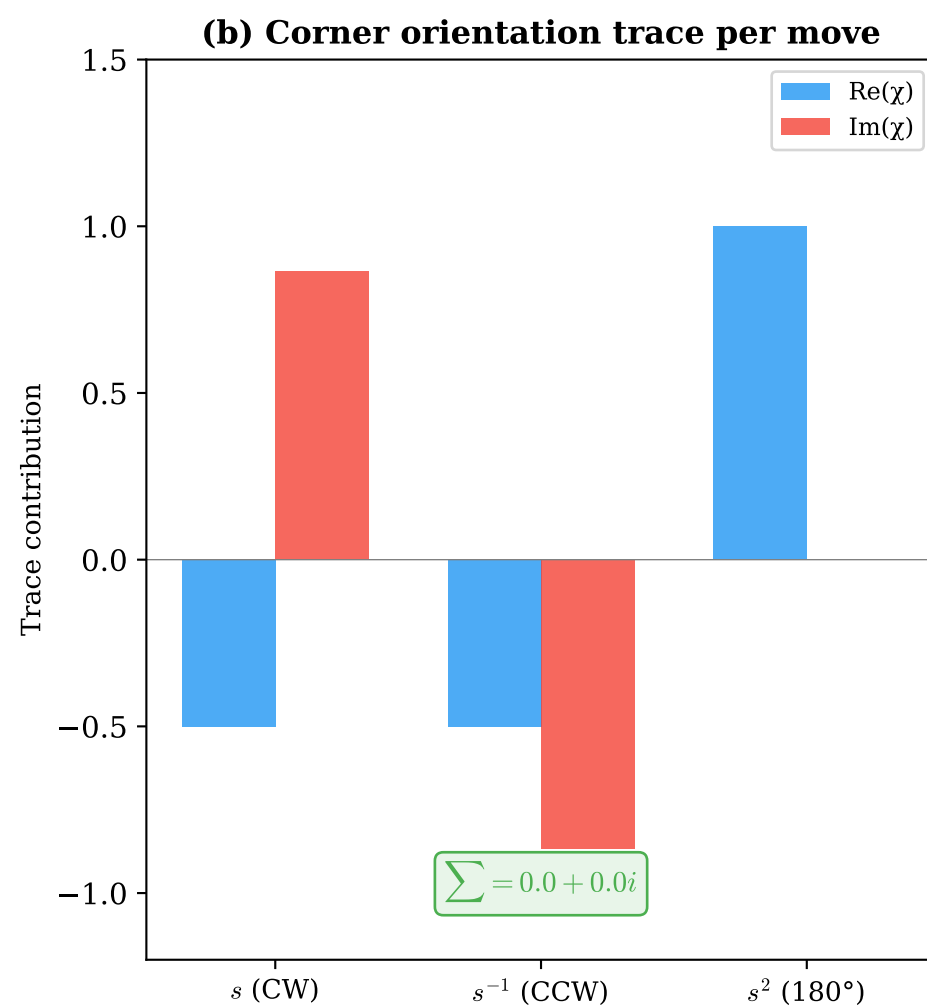
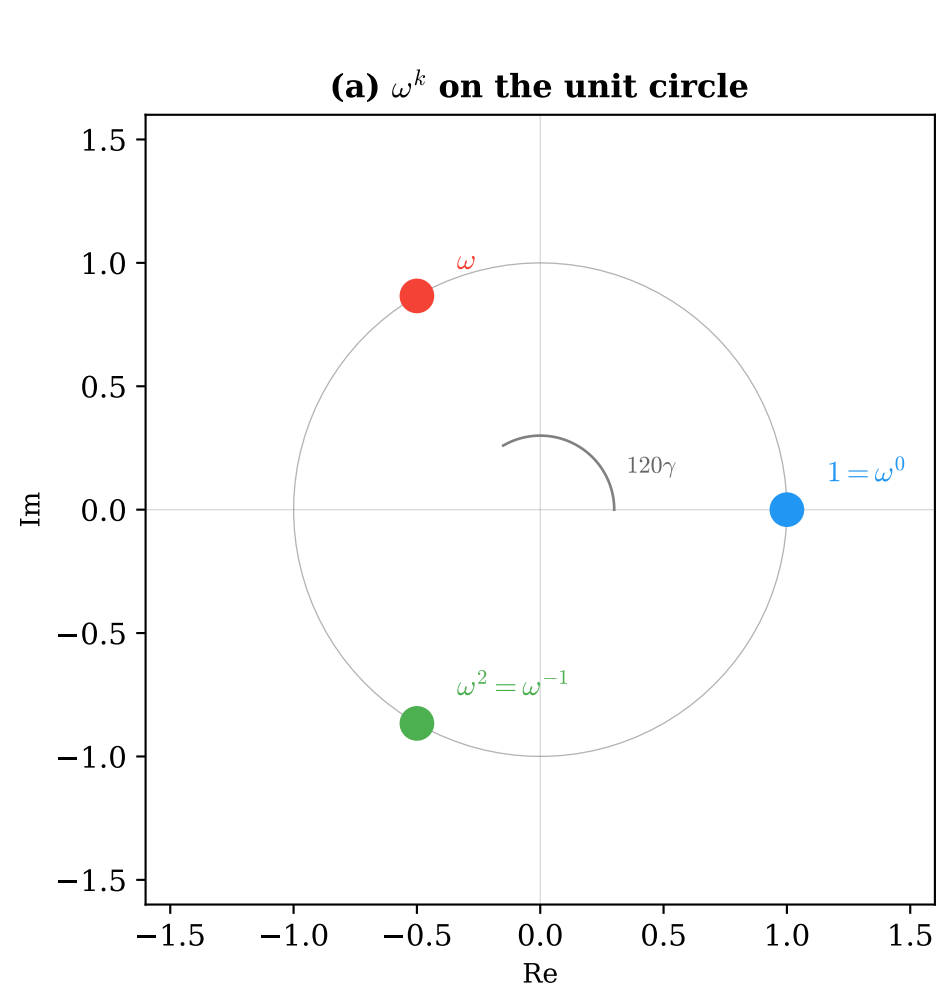




(c) Face-sum cancellation

Move	χ_{co}	Re	Im
s (CW)	ω	$\cos(120\gamma)$	$\sin(120\gamma)$
s^{-1} (CCW)	ω^2	$\cos(240\gamma)$	$-\sin(120\gamma)$
s^2 (180°)	1	1	0

$\omega + \omega^{-1} + 1 = 0$
Imaginary parts cancel $\Rightarrow$ face-sum $\in \mathbb{Z}$